

MarketLens

AI market research that shows its work. Every number links back to the tool call that measured it, and if the data fails it refuses to write rather than guess.

LIVE APP	themarketlens.streamlit.app
PROJECT PAGE	aarushkik.github.io/ReverieStockPlatform
SOURCE	github.com/aarushkik/ReverieStockPlatform

The problem

People are already making financial decisions on AI answers they cannot check. The failure is not that models are unintelligent. It is that when you read an answer, nothing tells you which numbers came from real data and which the model produced on its own. Every figure looks equally confident.

We found this in our own app first.

Run the terminal with the network off and it happily reported a market cap of exactly \$250 billion and a P/E of 24.5, both invented by a fallback path. One function derived a stock's daily percent change from the letters in its ticker symbol. Nobody wrote that to deceive. It arrived one reasonable looking except block at a time.

What it does

MarketLens runs AI research as a **workflow instead of a chat**. You choose what you want to know and it executes a pipeline of real tools over real market data while you watch: prices, indicators, fundamentals, news, sentiment. It then writes a memo in which every figure is a receipt. Hover any number to see the exact tool call that measured it, the provider, and how long ago.

Asking a model to cite sources is close to worthless, because it will attach a convincing citation to a number it invented. So MarketLens reads its own finished memo back, extracts every number, and checks each against what the tools actually returned.

STATUS	MEANING
verified	the number matches what was measured
uncited	a number with no source at all
unknown_fact	cites an id that does not exist
mismatch	cites something real but states a different number

The part people remember

Give it a symbol that does not resolve. The pipeline goes red, dependent steps are skipped, and no memo is produced at all. There is a working language model available and unused. It would rather say nothing than write around the gap.

HOW IT WORKS

Architecture

The workflow picks the tools, not the model. Steps run as a graph in dependency order, independent ones in parallel. The model never chooses what to call; it only reasons over data it did not select. That trades open ended agency for reproducibility, so a run can be replayed and audited.

Underneath sits a fact ledger. Every value a tool produces is registered with an id, a unit and its origin before the model sees anything. The model may only cite from that table, and the verifier checks each number in the finished memo against the fact it cited.

Measured results

WHAT	RESULT
Due diligence run, 6 steps, live data	2.5 s, 36 facts measured
Memo verification against the ledger	16 claims, 16 verified
Model instructed to invent 4 figures	refused all four
Data source unavailable	run halted, no memo written
Login risk model, held out	ROC AUC 0.966
Bot detection model, held out	ROC AUC 0.978
Test suite, offline	158 passing in ~1 s

Honesty work underneath

The verifier is only credible because the data layer stopped lying first. Every fabrication was removed: invented fundamentals, synthetic market movers, fabricated insider trades attributed to real named executives, and a model reporting a backtest accuracy it had never computed. Fetchers now return a measurement or raise.

Built with

Application	Python, Streamlit, Plotly
Market data	yfinance, Finnhub, Yahoo Finance RSS
Models	scikit-learn for login risk, bot detection and price direction
Language model	provider agnostic; Featherless, OpenAI, Anthropic, DeepSeek, Gemini
Testing	pytest, 158 tests, no network required

What's next

Run history with replay and diff, so the same question asked a week later shows exactly which facts moved. Portfolios that survive a refresh. And citation checking in every corner of the app, not just the workflow tab.

Try it in two minutes

Open the live app and choose **Create account**; there are no preset logins. Open Workbench, run Due diligence on AAPL, hover any figure in the memo. Then run it again with a symbol that does not exist.